

Particles as Topological Kinks

From the Phi-MDL Field to the Standard Model Spectrum

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UGP Physics Tutorial Series

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1 What Is a Particle in GTE?

The one-sentence answer

In the GTE framework, a particle is not a point object inserted into a field as a separate ingredient — it is a stable, localised *twist* in the Φ_{MDL} field itself, where the field transitions from one vacuum state to another without being able to unwind.

In the Standard Model, elementary particles are fundamental objects defined by their quantum numbers (charge, spin, colour). The SM writes down a Lagrangian and adds the particles as separate ingredients; their masses are free parameters fitted to experiment.

In **GTE (Generative Triple Evolution)**, there is only one object: the Φ_{MDL} **field**, a quantum field with internal symmetry group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ evolving on $\mathbb{R}^{3+1}$. Particles are not added separately — they emerge as localised excitations called **topological kink solitons** [?].

1.1 The Potential Landscape: Seven Degenerate Vacua

The Φ_{MDL} potential has exactly seven degenerate minima, labelled $w \in \{0, 1, 2, 3, 4, 5, 6\}$ — the elements of $\mathbb{Z}_7$. This is not a model choice: the $\mathbb{Z}_7$ symmetry is forced by **MDL (Minimum Description Length)** selection, the principle that identifies F_{21} as the uniquely shortest description consistent with the Standard Model derivability criterion [?].

At any point in space the field Φ_{MDL} lives in one of these seven minima. The ground state of the universe — the vacuum — sits uniformly at $w = 0$.

1.2 What a Kink Is

Definition: Topological Kink (BPS Soliton)

A **kink** is a field configuration that interpolates smoothly from one vacuum $w = a$ on the left to a different vacuum $w = b$ on the right as a function of position along one spatial axis. The integer $\Delta w = b - a \pmod{7}$ is the **winding number** of the kink.

Think of the seven vacua as seven towns on a circular road. A kink is a road segment connecting two consecutive towns. You cannot straighten it out without leaving the road — the topology forbids it. The kink is therefore *topologically protected*: it cannot decay into the vacuum by any continuous deformation of the field.

An everyday analogy: a twisted rubber band. Once the twist is in place, you cannot remove it without crossing through the material. The twist cannot “gradually untwist” while staying on the rubber. A kink in the Φ_{MDL} field is the same idea in a continuous field.

1.3 Why Kinks Are Stable

There are two independent reasons for kink stability.

Topological reason. The **winding number** w is an exact conserved charge of the $\mathbb{Z}_7$ field algebra (machine-certified: `gte_baryon_number_topological_charge`, **CatAL**) [?]. It cannot change value by any process that acts locally on the field.

Energetic reason. Kinks satisfying the **BPS condition** (explained in §??) saturate a lower bound on their energy set by topology alone. They are the *lowest-energy* configurations in their winding class. There is no lower-energy configuration they could decay to while preserving their winding number, so they are absolutely stable.

2 BPS Kinks: Energy, Mass, and Saturation

Key idea

BPS stands for **Bogomolny-Prasad-Sommerfield**. A BPS kink is one that saturates the *Bogomolny energy bound* — the minimum energy a field configuration with a given winding number must have. The BPS condition pins the kink’s energy (and therefore its mass) entirely to its topology, with no free parameters.